MIL-F-47250(MI)
26 July 1974
SUPERSEDING
MIS 16090A
26 June 1967

MILITARY SPECIFICATION

FABRIC, UPHOLSTERY, NYLON, COTTON, ACRYLIC FIBER, HONEYCOMB

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of upholstery fabric consisting of nylon, cotton, and acrylic fibers treated with a flame retardant acrylic resin or chlorinated rubber.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-P-1132	Packaging and Packing of Woolen, Worsted and Wool Blend (Synthetic Fiber, Cotton) Fabrics

FSC 8305

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T08:05Z
Check the source to verify that this is the current version before use.

STANDARDS

Federal

FED-STD-191	Textile Test Methods
FED-STD-595	Color

Military

MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Federal Trade Commission

Rules and Regulations Under the Textile Fiber Products Identification Act

General Services Administration, Federal Supply Service

FSSR 453	Fire Preventive Test Procedure for Airframe Materials (Flame Resistant Materials)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), a preproduction sample of fabric shall meet the requirements of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T08:05Z
Check the source to verify that this is the current version before use.

3.2 Material. The material shall consist of a nylon, cotton, and acrylic fiber woven honeycomb upholstery fabric coated with a flame retardant acrylic resin or chlorinated rubber.

3.3 Physical properties.

3.3.1 Width. The finished width of the fabric shall be not less than 54 inches (see 6.2).

3.3.2 Length. The roll length shall be 40 to 60 yards (see 6.2).

3.3.3 Weight. The finished weight of the fabric shall be 12 plus or minus 1.0 ounces per square yard.

3.3.4 Weave. The weave shall be a honeycomb type with 91.2 plus or minus 5 ends per inch and 80 plus or minus 5 picks per inch.

3.3.4.1 Warp. The warp shall be as follows:

a. 20/2 KP cotton.

b. 2/34 WC N4887 nylon.

c. 3/200 SD acrylic fiber.

3.3.4.2 Filling. The filling shall be as follows:

a. 2/34 WC N4887 nylon.

b. 20/2 KP cotton.

3.3.5 Fiber content (theoretical). The fiber content shall be in accordance with Table I:

Table I. Fiber content

Fiber	Fiber content, %
Nylon	46.6 ± 2.0
Cotton	48.7 ± 2.0
Acrylic fiber	4.7 ± 0.5

3.3.6 Fiber identification. Each roll shall be labeled, ticketed, or invoiced for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

3.3.7 Shrinkage. The shrinkage shall be not more than 5 percent after dry cleaning.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T08:05Z
Check the source to verify that this is the current version before use.

3.3.8 Odor. The odor of the finished fabric shall not be objectionable.

3.3.9 Color. The color of the finished fabric shall be green approximating Color Number 34092 of FED-STD-595.

3.3.10 Burning rate. The burning rate of the fabric shall be not more than 4 inches per minute. The fabric shall conform to the requirements of FSSR 453.

3.3.11 Appearance retention. The appearance retention of the fabric shall match a standard sample that has been mutually agreed upon by the supplier and the procuring activity.

3.4 Mechanical properties.

3.4.1 Abrasion resistance. The abrasion resistance shall be determined with a Taber Abraser using a CS-10 abrading wheel or approved equivalent with 1000 gram load for 500 cycles. There shall be no noticeable wear or change to the fabric.

3.4.2 Tensile strength. The tensile strength, before tear, shall be not less than 200 pounds per square inch (psi) in the warp direction and 100 psi in the filling direction.

3.5 Contamination. The fabric shall be clean and free from dirt, oils, foreign material, or other contaminants.

3.6 Workmanship. The workmanship shall be such as to insure a high quality product which is uniform and in conformance with this specification. The fabric shall be free of slubs, knots, pulls or other defects that could affect the unsability or appearance, in accordance with practices standard in the textile industry.

4. QUALITY ASURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample. The preproduction sample shall be prepared using the same methods proposed for the preparation of subsequent lots of material. Preproduction samples which do not meet the

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T08:05Z
Check the source to verify that this is the current version before use.

requirements of this specification shall be subject to rejection and return the supplier.

4.3 Classification of tests. The inspection and testing of the finished fabric shall be classified as follows:

a. Preproduction tests.

b. Quality conformance tests.

4.3.1 Preproduction tests. Preproduction tests shall be conducted only on the preproduction sample and shall consist of all the examinations and tests specified herein.

4.3.2 Quality conformance tests. Quality conformance tests for acceptance of the finished fabric shall consist of the following examinations and tests:

a. Width.

b. Weight.

c. Color.

d. Burning rate.

e. Contamination.

f. Workmanship.

4.3.2.1 Lot size. Lot size shall consist of all the finished fabric manufactured at one time from one run, forming part of one contract or order, and submitted for acceptance at the same time and place.

4.3.2.2 Sampling. Unless otherwise specified (see 6.2), sufficient fabric from one run, representative of each lot shall be selected at random for quality conformance testing. Each run shall be considered as a unit of product. Failure of sample to meet the quality conformance test requirements shall be cause for lot rejection.

4.4 Test methods.

4.4.1 Test conditions. Unless otherwise specified in the applicable test, all tests shall be conducted at plus 21 or minus 1 degrees Celsius (C) (plus 70 plus or minus 2 degrees Fahrenheit (F)) and at relative humidity of 65 plus or minus 2 percent.

4.4.2 Physical properties.

4.4.2.1 Width. The width shall be determined in accordance with FED-STD-191, Method 5020. The width shall be as specified in 3.3.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T08:05Z
Check the source to verify that this is the current version before use.

4.4.2.2 Length. The length shall be determined in accordance with FED-STD-191, Method 5010. The length shall be as specified in 3.3.2.

4.4.2.3 Weight. The weight shall be determined in accordance with FED-STD-191, Methods 5040 or 5041. The weight shall be as specified in 3.3.3.

4.4.2.4 Weave. The weave shall be determined by visual examinations and by count of ends and picks in accordance with FED-STD-191, Method 5050. The weave shall be as specified in 3.3.4.

4.4.2.5 Warp. The warp shall be determined by visual examination. The warp shall be as specified in 3.3.4.1.

4.4.2.6 Filling. The filling shall be determined by visual examination. The filling shall be as specified in 3.3.4.2.

4.4.2.7 Fiber content. The fiber content shall be determined theoretically. The fiber content shall be as specified in Table I.

4.4.2.8 Fiber identification. Each roll shall be examined for compliance with the Rules and Regulations Under the Textile Fiber Products Identification Act. Each roll not labeled in this act shall be cause for rejection. Fiber identification shall be as specified in 3.3.6.

4.4.2.9 Shrinkage. The shrinkage shall be determined in accordance with FED-STD-191, Method 5580. The shrinkage shall be as specified in 3.3.7.

4.4.2.10 Odor. The odor shall be determined by actually smelling the finish fabric. The odor shall be as specified in 3.3.8.

4.4.2.11 Color. The color of the finished fabric shall be determined visually be comparison with the Color Method Number 34092 of FED-STD-595. The color shall be as specified in 3.3.9.

4.4.2.12 Burning rate. The burning rate shall be determined in accordance with FED-STD-191, Method 5906. The burning rate shall be as specified in 3.3.10.

4.4.2.13 Appearance retention. The appearance retention of fabric shall be determined in accordance with FED-STD-191, Method 5310. The appearance retention shall be as specified in 3.3.11.

4.4.3 Mechanical properties.

4.4.3.1 Abrasion resistance. The abrasion resistance shall be determined in accordance with FED-STD-191, Method 5306. The abrasion resistance shall be as specified in 3.4.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T08:05Z
Check the source to verify that this is the current version before use.

4.4.3.2 Tensile strength. The tensile strength shall be determined in accordance with FED-STD-191, Method 5100. The tensile strength shall be as specified in 3.4.2.

4.5 Contamination. The contamination shall be determined by visual examination. The contamination shall be as specified in 3.5.

4.6 Workmanship. The workmanship shall be determined visually. The workmanship shall be as specified in 3.6.

4.7 Preservation, packaging, packing, and marking. The preservation, packaging, packing and marking shall be examined for compliance with Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified (see 6.2), preservation, packaging, packing, and marking shall be as specified herein.

5.1.1 Preservation and packaging. Preservation and packaging shall be Level A or C, as specified (see 6.2) in accordance with the applicable requirements for Class 3 of PPP-P-1132.

5.1.2 Packing. Packing shall be Level A, B, or C, as specified (see 6.2) in accordance with the applicable requirements of Class 3 of PPP-P-1132.

5.1.3 Marking. In addition to any special marking required by the contract or purchase order, shipments shall be marked in accordance with the requirements of PPP-P-1132.

6. NOTES

6.1 Intended use. The woven honeycomb fabric is intended for use as aircraft seating upholstery.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether a preproduction sample is required, and, if so, pertinent details (see 3.1 and 4.3.1).

c. Sampling plan if other than specified (see 4.3.2.2).

d. Requirements for preservation, packaging, packing, and marking, if different (see 5.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T08:05Z
Check the source to verify that this is the current version before use.

e. Width and length required, if different (see 3.3.1 and 3.3.2).

6.3 Supersession data. This specification includes the requirements of Missile Interim Specification MIS-16090A, dated 26 June 1970.

Custodian:
ARMY - MI

Preparing Activity:
ARMY - MI
Project No. 8305-A319

☆ U. S. GOVERNMENT PRINTING OFFICE: 1974-603-109/813

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T08:05Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

S/N 0102-014-1802

☆ U.S. GOVERNMENT PRINTING OFFICE: 1972—714-863/3404 2-1

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T08:05Z
Check the source to verify that this is the current version before use.